

Forecasting LA Home Values

MGMTMSA 437 Forecasting and Time Series Analysis · Final project · Oct 7, 2026 · @Neil

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Executive summary

LA home values rose 5.6% a year from 2000 to 2026, then stalled. In August 2026 the typical LA-metro home was worth \$952,601, 1.3% below its January 2025 peak. We asked two questions: which forecasting method predicts LA home values best, and do mortgage rates improve the forecast?

Data. The Zillow Home Value Index (ZHVI) for the LA metro, January 2000 to August 2026 (320 months), plus 499 LA-metro neighborhoods and the 30-year fixed mortgage rate from FRED.

Method. We compared 11 models, from a naive drift baseline to LSTM and GRU networks. Three of them use extra information: ARIMAX with mortgage rates, a VAR, and a dynamic factor model built from 405 neighborhoods. Every model was scored two ways: a blind 80/20 test and a rolling-origin test (18 forecast origins, 1 to 12 months ahead).

Findings

1. ARIMAX with the mortgage rate lagged 24 months had the lowest rolling error: 2.67% on average over 1 to 12 months, slightly ahead of ARIMA at 2.77%. The rate signal is real but weak ($p = 0.16$).
2. At 12 months ahead, no model beat a naive drift (4.92% error vs 4.98% for ARIMAX). Zillow smooths this index, so 1-month forecasts are easy (about 0.1% error) and say little about model quality.
3. Machine learning and deep learning did not beat simple models. LightGBM won the single blind test (2.2%) but placed eighth of 11 in the rolling test, which shows how one test window can mislead.
4. The 2022 rate shock hit expensive LA County neighborhoods hardest, and the market is still split. Over the last year the coastal high end (Beverly Hills, Santa Monica, coastal Orange County) led again, while central LA (MacArthur Park, Koreatown, Downtown, Westlake) and fire-hit Pacific Palisades fell.

5. One common factor explains 84% of neighborhood-level movement, yet separate ARIMA models per neighborhood forecast better (2.8% error) than the shared-factor model (4.1%).

Outlook. Our models point to a 4.6% to 4.8% rise for the LA metro by August 2027, to about \$998,000. The 80% interval is wide, from -6% to +17%. Forecasts for individual neighborhoods range from -2.1% to +12.0%.

Purpose and motivation

LA is one of the most expensive housing markets in the US, and a home is the largest asset most households own. Between 2021 and 2023 the 30-year mortgage rate jumped from 2.7% to 7.6% (monthly averages). That makes LA a good test case: did a shock that large make home values harder to forecast, and do rates themselves help predict where prices go?

A good forecast matters to several groups. Buyers and sellers time transactions around it. Lenders and investors price risk with it. City and county planners use it to project property-tax revenue and affordability.

The project has three goals:

1. **Forecast** LA metro home values 12 months ahead, with an honest error band.
2. **Compare methods.** Test whether machine learning and deep learning beat classical time-series models on a monthly housing index, under a fair evaluation.
3. **Explain the shock.** Measure how much mortgage rates add to the forecast, and which LA neighborhoods the 2022 rate shock hit hardest.

Data

All three series are monthly, public, and run from January 2000 to August 2026.

Series	Source	Coverage	Role
ZHVI, LA metro (Los Angeles-Long Beach-Anaheim)	Zillow Research	320 months	Forecast target

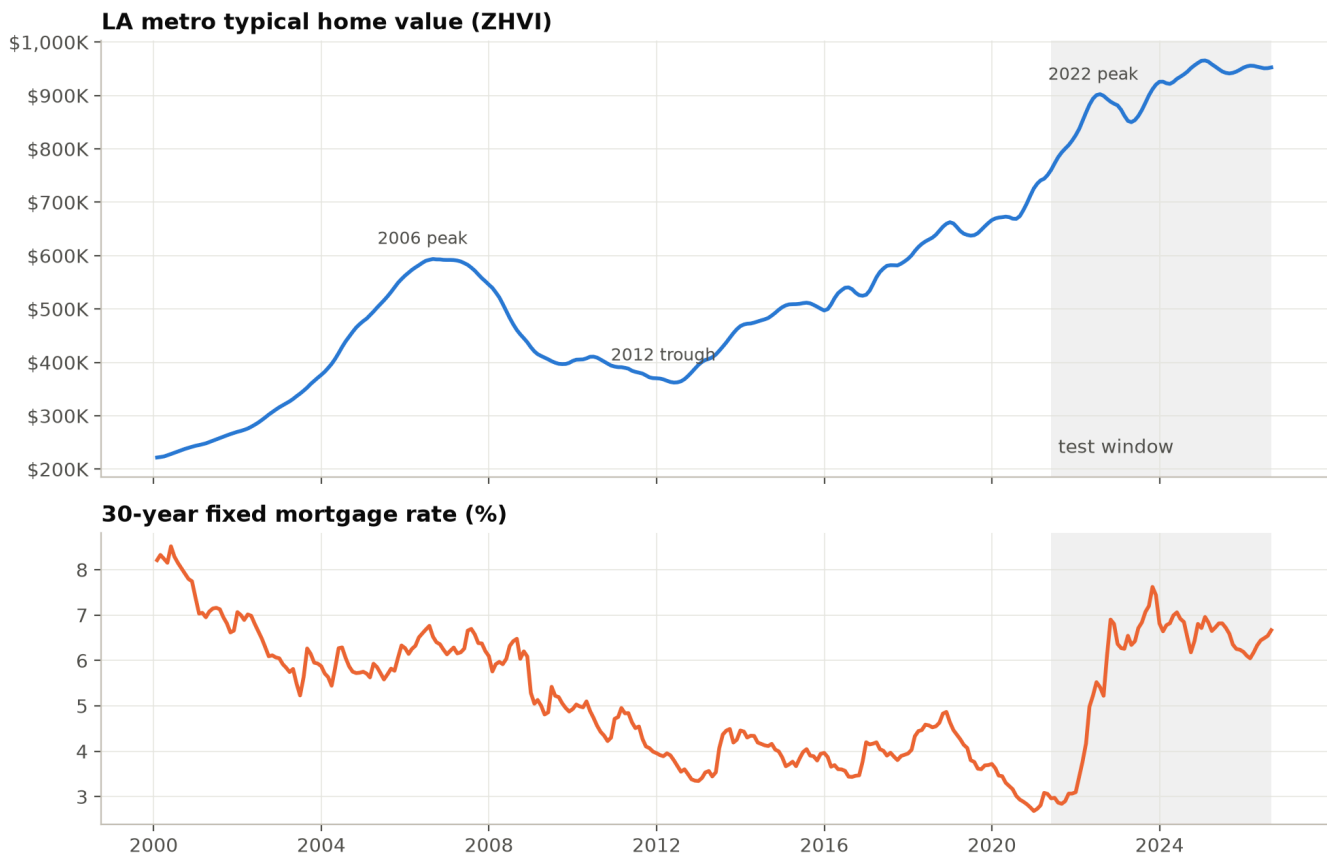
Series	Source	Coverage	Role
ZHVI, LA-metro neighborhoods	Zillow Research	499 neighborhoods (361 LA County, 138 Orange County); 405 with full 2000–2026 history	Neighborhood analysis, factor model
30-year fixed mortgage rate (<code>MORTGAGE30US</code>)	FRED, Freddie Mac	Weekly, averaged to monthly	Explanatory variable

ZHVI measures the typical value of homes in the 35th to 65th percentile range, covering single-family homes and condos. We use Zillow's smoothed, seasonally adjusted version. That choice has two consequences we come back to later: the series has almost no seasonality left, and month-to-month changes are very persistent.

Neighborhoods use Zillow's own neighborhood boundaries. The shock and trend analysis uses the 496 neighborhoods with data since 2020; forecasting uses the 405 with complete history. Zillow's full neighborhood file is 103 MB, so we ship only the LA subset (2.7 MB) and link the source. Map positions are neighborhood centroids geocoded from OpenStreetMap.

Exploratory analysis

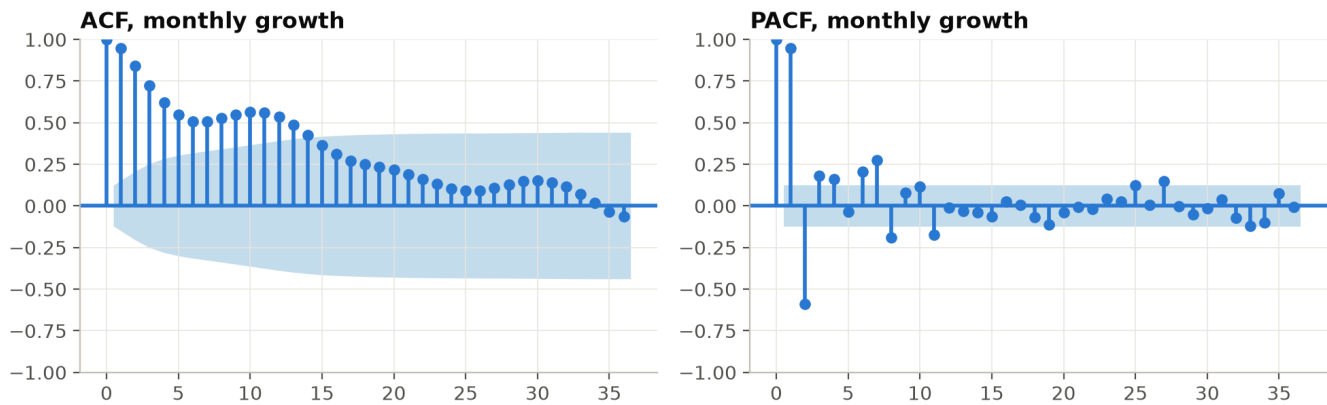
LA home values are not a steady upward line. They went through one full boom and bust, a long recovery, a pandemic surge and a stall once rates jumped.



- **Long-run growth:** 5.6% a year from January 2000 to August 2026.
- **2006 to 2012 bust:** values fell 37.2% from the May 2006 peak to the February 2012 trough, and did not regain the 2006 peak until December 2017.
- **2020 to 2022 boom:** with mortgage rates near 3%, values rose 34% from February 2020 to the 2022 peak.
- **Since 2022:** rates more than doubled. Values dipped in 2023, recovered to a new peak of \$965,634 in January 2025, and have drifted sideways since.

Stationarity

Log home values clearly have a unit root (ADF $p = 0.56$), so every model works with differences or growth rates. Monthly growth is borderline (ADF $p = 0.16$). Because Zillow smooths the index, monthly growth is highly persistent: its autocorrelation is 0.95 at one month and still about 0.5 at twelve. That persistence means a model can forecast next month almost perfectly just by repeating this month, so a 1-month error tells us little. We judge models on 6- and 12-month horizons instead.



Do mortgage rates lead home values?

In the training period (2000 to April 2021), the 12-month change in mortgage rates lines up with LA home-value growth over the following two years:

Months after the rate change	Correlation with year-over-year home-value growth
0	+0.33
6	+0.11
12	-0.14
18	-0.29
24	-0.35

In the same month the correlation is positive, because rates and prices both rise in a strong economy. The negative effect shows up 12 to 24 months later. That delay shapes the ARIMAX model in the next section.

Models and why we chose them

We picked 11 models to cover four families: simple baselines, classical time series, machine and deep learning, and models that bring in outside information. Every model forecasts log home values. The machine-learning and neural models predict monthly growth and roll forward month by month, because trees cannot extrapolate a price level they have never seen.

Model	Family	Inputs	Why it is here
Naive drift	Baseline	LA's own average monthly growth	The bar every other model must clear
ARIMA(1,1,2) with drift	Classical	LA history only	Order chosen by AIC (<code>auto_arima</code>) on the training set
Holt, damped trend	Classical	LA history only	Exponential smoothing. No seasonal term, since the series is already seasonally adjusted
Prophet	Classical	LA history only	Piecewise trend with automatic changepoints
XGBoost, LightGBM	Machine learning	Growth lags 1 to 12, 3/6/12-month moving averages	Can capture nonlinear momentum
LSTM, GRU	Deep learning	Previous 24 months of growth	Learn sequence patterns. 32 hidden units, early stopping on the last 10% of training data
ARIMAX	Outside information	LA history + mortgage rates	Tests whether rates add forecasting power
VAR	Outside information	LA growth + monthly rate changes, 11 lags	Forecasts rates and home values together, so it needs no assumed rate path
Factor VAR	Outside information	LA growth + 2 factors from 405 neighborhoods	Big-data idea: let many neighborhoods inform the metro forecast

ARIMAX design

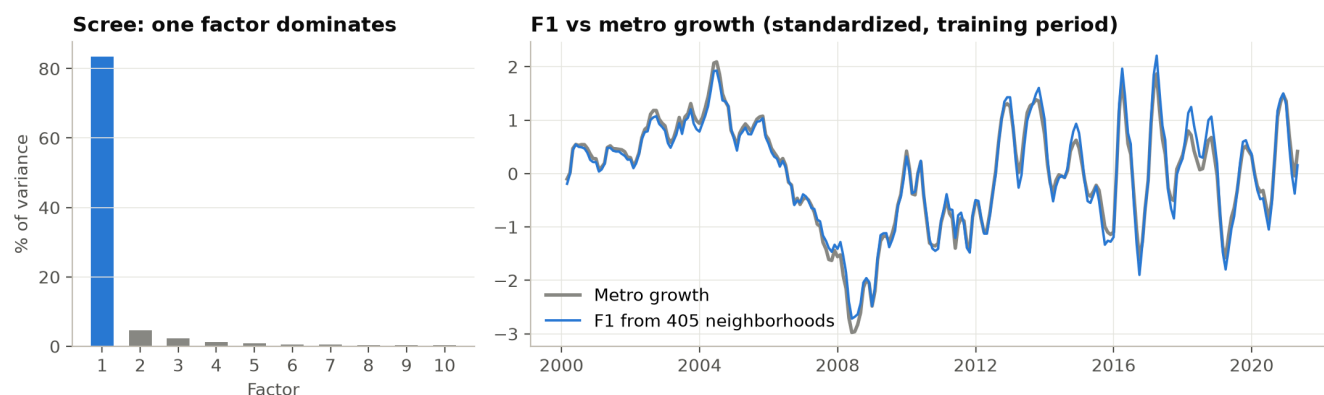
The exploratory analysis showed rates affect home values with a long delay, so ARIMAX models monthly growth as ARMA(1,2) errors plus the 12-month change in the mortgage rate lagged L months. We tried $L = 12, 15, 18, 21$ and 24 on the training data, and AIC picked 24.

The estimated coefficient is -0.063 (standard error 0.045 , $p = 0.16$). A 1-point rise in rates over a year lowers monthly growth by about 0.06 points two years later, roughly 0.75% of home value over the following year. The 12-month rate change peaked at 3.8 points in October 2022, which implies a drag of roughly 3% on home values spread over 2024. The effect has the expected sign and size, but it is not statistically significant, and adding the rate improves AIC only slightly. It is weak evidence, not proof.

A 24-month lag has a practical advantage: a 12-month forecast uses only rates we have already observed, so ARIMAX needs no guess about future rates. In the 64-month blind test it does eventually need future rates. There we show two versions: one that holds the last rate flat (a genuine forecast) and one that uses the rates that actually happened (a conditional benchmark, not a real forecast).

Dynamic factor model

The 405 neighborhoods with full history move together. One principal component of neighborhood growth explains 83.5% of their variance in the training period and tracks metro growth almost exactly (correlation 0.99). The second and third factors add 4.6% and 2.3%. The factor VAR uses the first two factors, re-estimated at every forecast origin on data up to that date, so no future information leaks in.



Evaluation design

We score every model out of sample in two ways, because a single train/test split can reward luck.

1. **Blind 80/20 test.** Train on January 2000 to April 2021 (256 months) and forecast May 2021 to August 2026 (64 months) in one shot, with no updates. This window contains the whole rate shock, which makes it a hard test.

2. Rolling-origin test. Starting at April 2021 and moving forward 3 months at a time, refit every model on all data up to that point and forecast the next 12 months. That gives 18 origins, from April 2021 to July 2025. Every model sees exactly the same information at every origin.

Model orders (ARIMA, ARIMAX lag, VAR lags) are chosen once on the training set. At each rolling origin only the parameters are re-estimated.

Metrics. In the blind test we report MAE and RMSE in dollars, MAPE in percent, and directional accuracy (the share of months where the forecast gets the direction of change right). In the rolling test we report mean absolute percentage error at 1, 3, 6 and 12 months ahead, and the average over all 12 horizons. That average is our main ranking metric.

Results: LA metro

ARIMAX and ARIMA forecast best in the rolling test, but the margins are small, and a naive drift is just as good 12 months out.

Rolling-origin test (main result)

Mean absolute % error across 18 forecast origins, sorted by the 1 to 12 month average:

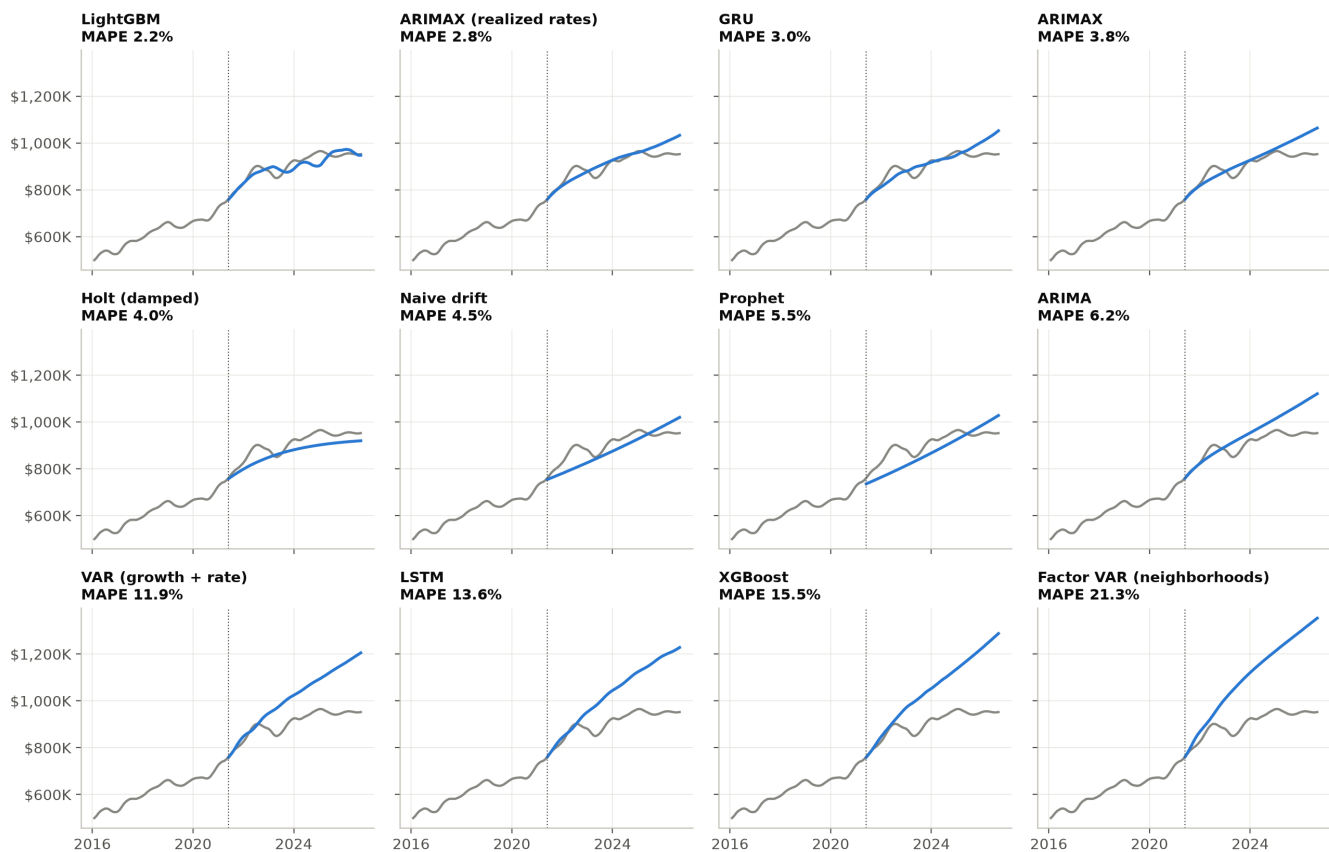
Model	1 month	3 months	6 months	12 months	Average, 1–12
ARIMAX	0.09	0.80	2.45	4.98	2.67
ARIMA	0.09	0.78	2.53	5.37	2.77
VAR	0.15	0.94	2.56	5.74	2.93
LSTM	0.13	0.89	2.95	5.58	3.02
GRU	0.11	0.93	2.99	5.86	3.18
Naive drift	0.69	1.93	3.49	4.92	3.27
Holt, damped	0.29	1.15	3.04	6.34	3.39
LightGBM	0.22	1.23	3.32	6.97	3.59
Factor VAR (neighborhoods)	0.16	1.15	3.56	6.60	3.64
XGBoost	0.25	1.25	3.40	6.82	3.65
Prophet	5.00	5.77	6.73	8.62	6.89

What the table shows:

- **Rates help, but only at longer horizons.** ARIMAX and ARIMA are tied through 3 months. By 12 months ARIMAX is 0.4 points better (4.98% vs 5.37%), which is where a 24-month rate lag should matter.
- **Nothing beats naive drift at 12 months.** A model that simply extends LA's long-run growth is as accurate a year out as anything we built. The fitted models earn their keep at 1 to 6 months.
- **Machine learning and deep learning don't win.** LSTM and GRU land mid-table. LightGBM and XGBoost rank eighth and tenth, with the neighborhood factor VAR between them. With about 250 monthly observations there isn't much nonlinear structure to learn.
- **Prophet is off from the first month.** It fits a smooth trend curve and doesn't anchor to the latest value, so it starts about 5% off. That's a poor fit for a smoothed index whose next value is almost always close to the last one.

Blind 80/20 test

Blind test: forecasts from May 2021 (blue) vs actual (gray)



Model	MAE (\$)	RMSE (\$)	MAPE (%)	Direction right (%)
LightGBM	20,365	25,872	2.2	55
ARIMAX (realized rates)*	25,845	33,834	2.8	66
GRU	27,185	36,994	3.0	66
ARIMAX	34,948	46,536	3.8	66
Holt, damped	37,033	41,154	4.0	66
Naive drift	40,259	46,423	4.5	66
Prophet	49,076	55,419	5.5	64
ARIMA	58,112	74,679	6.2	66
VAR	110,794	134,068	11.9	66
LSTM	126,961	153,778	13.6	66
XGBoost	144,235	175,061	15.5	66
Factor VAR (neighborhoods)	197,913	231,503	21.3	66

*Uses the mortgage rates that actually happened, so it is a benchmark, not a real forecast.

The blind test ranks the models very differently, for three reasons:

- **Momentum models overshoot.** The training period ends in April 2021, in the middle of the pandemic boom. VAR, LSTM and XGBoost extend that boom, miss the 2022 stall, and end the test window roughly 25% to 35% too high; the neighborhood factor VAR overshoots even further.
- **LightGBM's win is mostly luck.** It is the only model whose forecast bends, and it happened to bend in the right years. In the rolling test, where it gets 18 tries instead of one, it ranks eighth.
- **Rates explain part of the stall.** ARIMAX with the realized rates (2.8%) beats ARIMAX holding rates flat (3.8%). Knowing the rate path would have helped, but nobody knew it in April 2021.

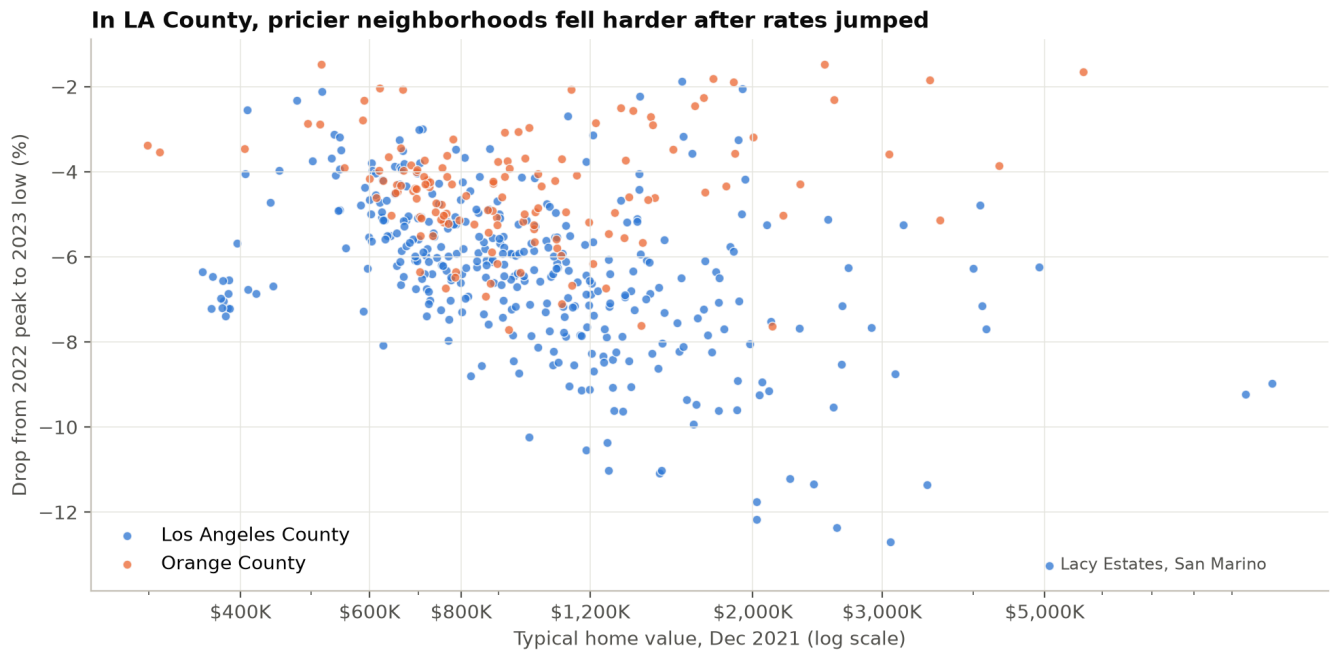
Directional accuracy is close to useless here. LA values rose in 42 of the 64 test months (66%), so any model that always predicts "up" scores 66%. LightGBM scores lower precisely because it is the only model that predicts some declines.

Neighborhoods: what the metro average hides

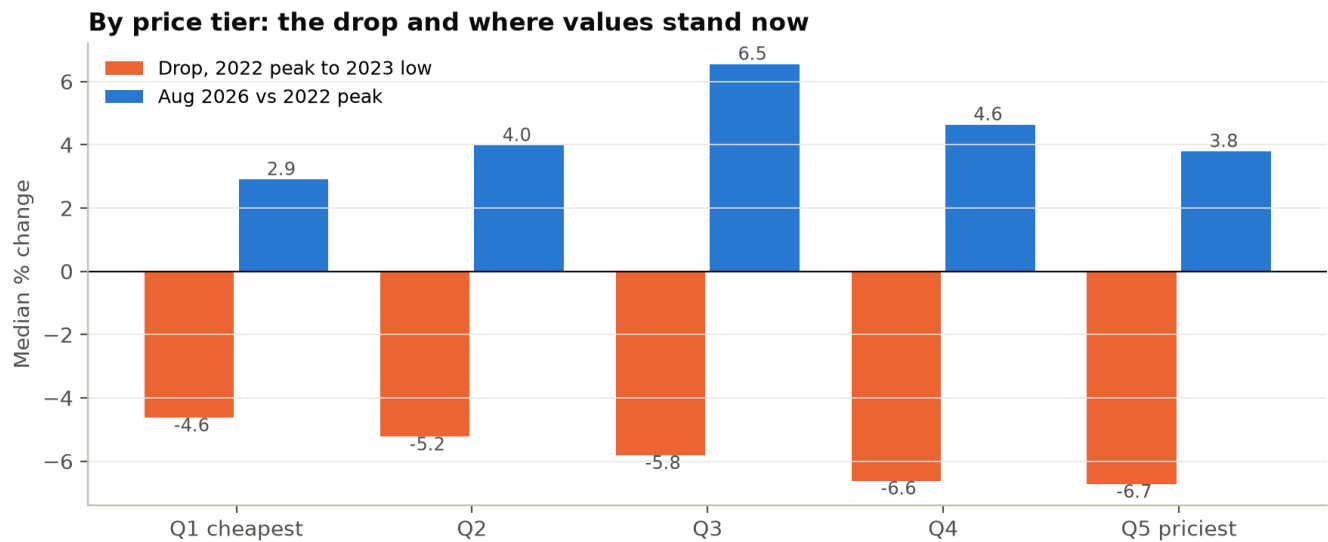
The metro average hides a split market. Four years after rates jumped, 152 of 496 neighborhoods are still below their 2022 peak, while several Irvine neighborhoods are more than 35% above theirs.

The 2022 rate shock, neighborhood by neighborhood

For each neighborhood with data since 2020 we measured the drop from its 2022 peak to its low point between mid-2022 and 2023, and where it stands today compared with that peak.



In LA County, the more expensive the neighborhood, the harder it fell (correlation between log price and the drop: -0.45). The five biggest drops were all in San Marino, led by Lacy Estates (-13.3%). A buyer of a \$2 million home needs a much larger loan, so a jump from 3% to 7% hits that buyer harder in dollars. In Orange County there is no such pattern (correlation $+0.10$).



	LA County	Orange County
Boom, Feb 2020 to 2022 peak	+32.5%	+37.9%
Drop, 2022 peak to 2023 low	-6.3%	-4.3%
August 2026 vs 2022 peak	+1.2%	+15.3%

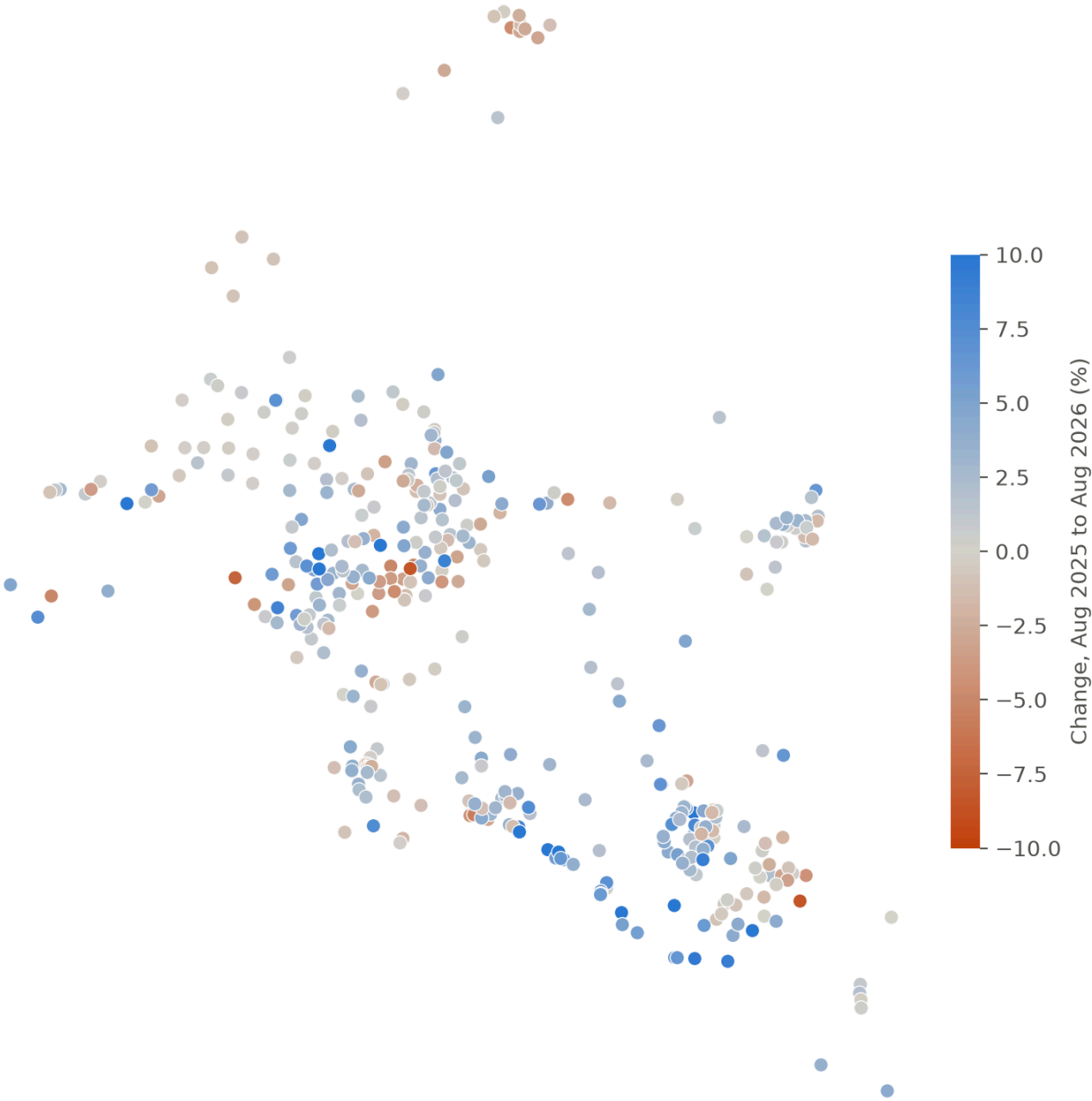
Medians across neighborhoods.

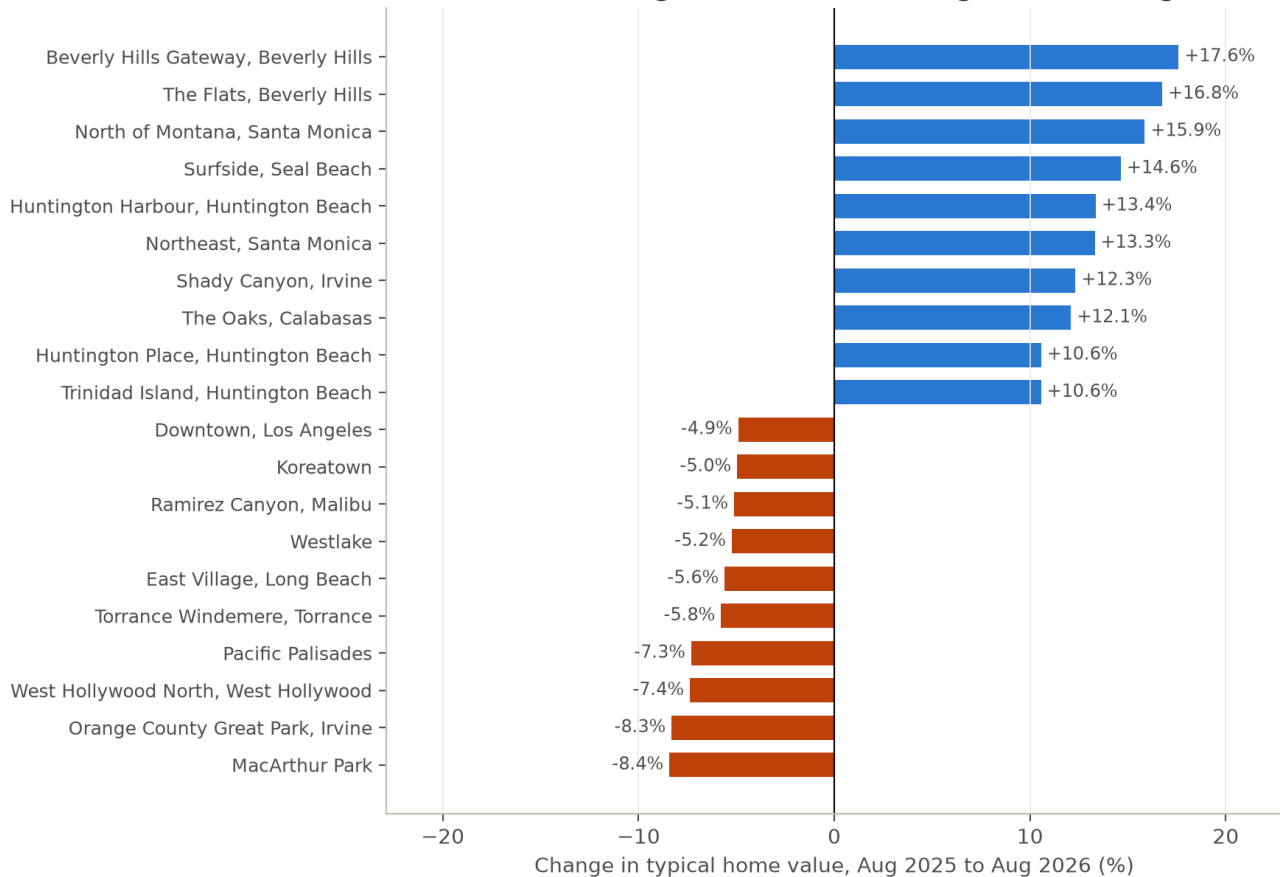
The neighborhoods furthest below their 2022 peak are central and urban: SoFo in West Hollywood (−30.4%), MacArthur Park (−24.0%), Arlington Heights (−19.4%), Hollywood (−17.9%) and Downtown Santa Monica (−17.7%). These areas have a high share of condos. One hypothesis is that condos recovered more slowly than single-family homes; Zillow’s separate condo index could test it.

Which neighborhoods are rising and falling now

Over the last 12 months (August 2025 to August 2026), 220 neighborhoods rose more than 2%, 216 were flat and 60 fell more than 2%.

LA-metro neighborhoods, last 12 months: 220 rising, 216 flat, 60 falling



The 10 fastest-rising and 10 fastest-falling LA-metro neighborhoods

	Rising	Flat	Falling
LA County	133	174	51
Orange County	87	42	9
City of Los Angeles (part of LA County)	22	56	18

The pattern has flipped since 2023. The high end that fell hardest is now leading: Beverly Hills Gateway (+17.6%), The Flats in Beverly Hills (+16.8%), North of Montana in Santa Monica (+15.9%) and coastal Orange County (Surfside +14.6%, Huntington Harbour +13.4%). The weakest neighborhoods are central LA (MacArthur Park -8.4%, Westlake -5.2%, Koreatown -5.0%, Downtown -4.9%), West Hollywood North (-7.4%) and Pacific Palisades (-7.3%), where the January 2025 fire, not the market, drives the decline. Only 22 of the 96 City of LA neighborhoods are rising.

An interactive version of this map, with every neighborhood's value, trend, 2022 drop, forecast and price history, accompanies this report.

Forecasting every neighborhood: shared factor or separate models?

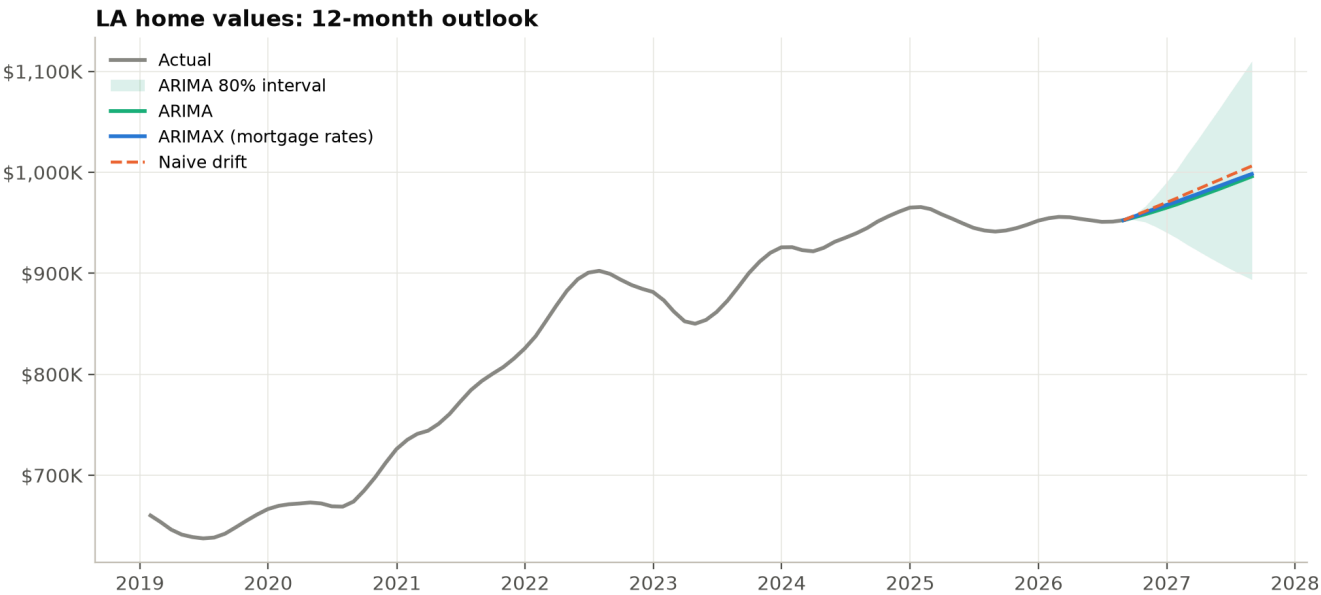
We forecast all 405 neighborhoods with full history 12 months ahead from 5 yearly origins (April 2021 to April 2025) with three methods:

Method	Average error, 1-12 months	Error at 12 months	Best for (% of cases)
ARIMA per neighborhood	2.84%	5.87%	41%
Naive drift per neighborhood	3.53%	6.89%	28%
Factor model (2 shared factors)	4.08%	8.20%	31%

Separate ARIMA models win. The factor model loses even though one factor explains 84% of neighborhood movement. The other 16% is each neighborhood's own trend, like the central-LA slump or the Irvine climb, and a two-factor model throws it away. Over 12 months those local trends add up to most of the forecast error. For the metro series the factor VAR also failed to beat plain ARIMA (3.64% vs 2.77%), because a factor that tracks the metro almost exactly carries no new information about it.

Outlook: September 2026 to August 2027

Our best model, ARIMAX, forecasts the typical LA-metro home at \$998,391 in August 2027, up 4.8% from \$952,601 today. The forecasts agree closely, but the uncertainty is wide.

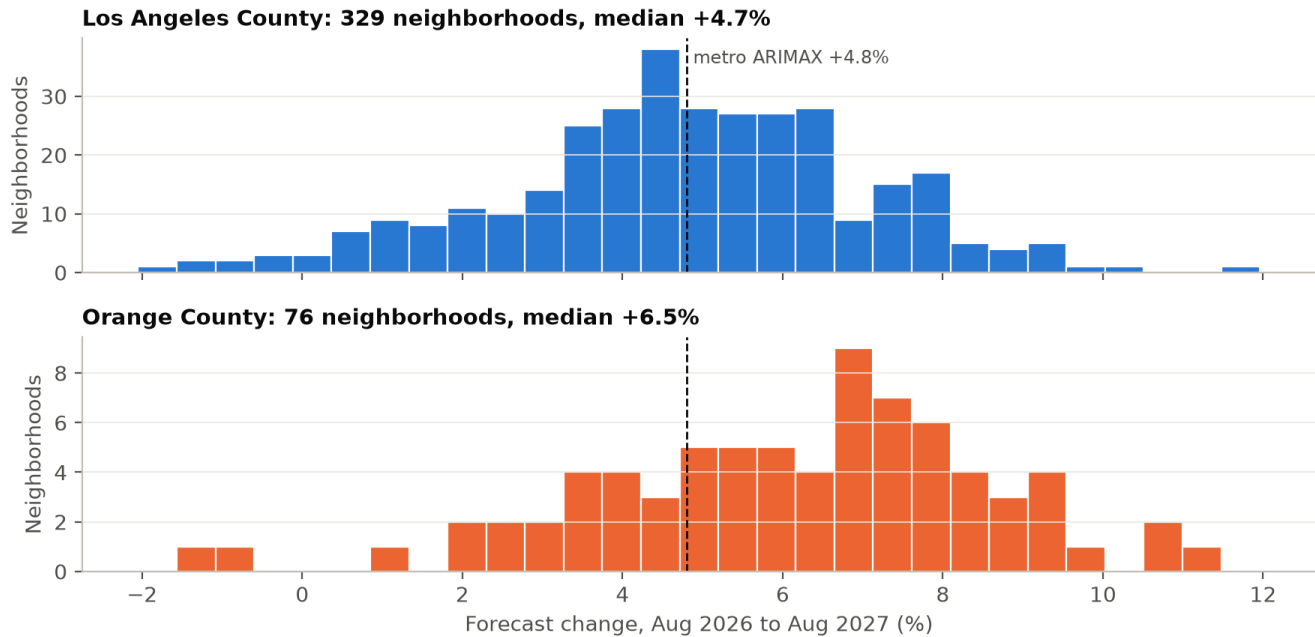


Model	August 2027 forecast	Change from August 2026
ARIMAX (mortgage rates)	\$998,391	+4.8%
ARIMA	\$996,023	+4.6%
Naive drift	\$1,006,266	+5.6%
ARIMA 80% interval	\$893,893 to \$1,109,822	-6.2% to +16.5%

Because of the 24-month lag, the next 12 months of the ARIMAX forecast depend on rate changes from September 2023 to August 2025, all already observed. Rates set in 2026 will first show up in 2028. The interval is the honest headline: the model gives a 10% chance of a fall of more than 6% and a 10% chance of a rise of more than 16%.

By neighborhood

Neighborhood outlook (ARIMA per neighborhood)



The median neighborhood forecast is +5.1%, and 80% of neighborhoods fall between +2.1% and +7.9%. Orange County neighborhoods are forecast to rise more (median +6.5%) than LA County neighborhoods (median +4.8%). Ten of 405 are forecast to fall, led by East Agoura in Agoura Hills (−2.0%). These forecasts mostly extend each neighborhood's recent trend, so they cannot see one-off events: Pacific Palisades' +5.7% forecast, for example, is a mechanical rebound after the fire, not a market signal.

Takeaways

- 1. For a monthly housing index, simple beats complex.** ARIMA-family models won the fair test. LSTM, GRU, XGBoost and LightGBM never beat them, and a naive drift was just as good a year out. With 250 smooth, persistent monthly observations, extra flexibility mostly fits noise.
- 2. How you test decides who wins.** One blind split crowned LightGBM. Eighteen rolling origins ranked it eighth. A forecast comparison built on a single test window, or on 1-month-ahead errors of a smoothed index, can make almost any model look good.
- 3. Mortgage rates matter, slowly and modestly.** Their effect on LA home values shows up about two years later and improves 12-month forecasts by about 0.4 points. That is useful for planning, but the statistical evidence is weak.

4. **The 2022 shock split LA by price and place, and the split has shifted.** Expensive LA County neighborhoods fell hardest in 2023, yet over the last year the coastal high end led again while central LA and fire-hit Pacific Palisades fell. Orange County is well above its 2022 peak; LA County barely is. The metro average describes almost no actual neighborhood.
5. **One shared factor drives 84% of the movement, but the rest is what matters for a forecast.** Local trends add up over a year, so each neighborhood needs its own model.

For a buyer or lender, the practical message is: expect about +5% for the typical LA home over the next year, but with a range from -6% to +16%. Look at the specific neighborhood, not the metro number.

Limitations and next steps

- **Smoothed data.** Zillow's smoothed, seasonally adjusted index hides month-to-month noise and seasonality. Raw ZHVI or the Case-Shiller index would give a harder test.
- **One overlapping test period.** The 18 rolling origins share data, so their errors are correlated. A Diebold-Mariano test would show whether ARIMAX's 0.1-point lead over ARIMA is statistically meaningful. We expect it is not.
- **One explanatory variable.** We used only mortgage rates. Employment, income, housing permits, inventory and rents (Zillow ZORI) are natural additions to the ARIMAX or VAR.
- **Neural models were lightly tuned.** One architecture, no hyperparameter search. Better tuning might narrow the gap, though the data size limits them.
- **The factor model used two factors and no neighborhood-specific dynamics.** A version that adds a separate AR term per neighborhood on top of the shared factors would keep the local trends we lost.
- **Neighborhood forecasts ignore one-off events** such as the 2025 fires and new development.

Appendix: reproducibility

Everything runs from one Python notebook, top to bottom, in about 4 minutes on a laptop.

File	Contents
<code>la_home_value_forecasting.ipynb</code>	Full analysis: data, EDA, models, both tests, neighborhood section, outlook

File	Contents
<code>la_home_value_forecasting.py</code>	The same code as a plain script
<code>data/Metro_zhvi.csv</code>	Zillow metro ZHVI (all US metros)
<code>data/la_nbhd_zhvi.csv</code>	LA-metro subset of Zillow's neighborhood ZHVI (499 neighborhoods)
<code>data/la_nbhd_centroids.csv</code>	Neighborhood map positions, geocoded from OpenStreetMap
<code>data/MORTGAGE30US.csv</code>	FRED 30-year mortgage rate, weekly
<code>figures/</code>	All charts plus result tables as CSV
<code>requirements.txt</code>	Package versions (pandas, statsmodels, pmdarima, prophet, xgboost, lightgbm, torch)

The full neighborhood-level ZHVI file is 103 MB, so it is not included. Download it from [Zillow Research](#) (ZHVI, all homes, mid tier, smoothed and seasonally adjusted, by neighborhood).

Sources: [Zillow Home Value Index](#) · [FRED, 30-Year Fixed Rate Mortgage Average \(MORTGAGE30US\)](#) · Hyndman and Athanasopoulos, *Forecasting: Principles and Practice* · Stock and Watson (2002), "Forecasting Using Principal Components From a Large Number of Predictors," *JASA* 97(460).